

PROJECT REPORT

CircuitGuard : AI-Driven Automated Optical Inspection (AOI) System

Submitted By

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Under the Guidance of **Ms. Pranaya**

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ABSTRACT

The electronics manufacturing industry relies heavily on the **structural integrity of Printed Circuit Boards (PCBs)**, which serve as the backbone of modern electronic devices. As component density increases and circuit features miniaturize, traditional **Quality Assurance (QA)** methods face significant challenges. **Manual visual inspection** is labor-intensive, subjective, and prone to error due to inspector fatigue, while industrial **Automated Optical Inspection (AOI)** systems are often cost-prohibitive and lack the flexibility to adapt to diverse production runs without complex recalibration.

CircuitGuard addresses these critical gaps by engineering a **hybrid, software-centric AOI solution**. This project unifies deterministic **Computer Vision** algorithms for precise defect localization with probabilistic **Deep Learning** models for robust classification. A key innovation of the system is the **Auto-Template Matching Engine**, which eliminates manual calibration by autonomously identifying the correct reference PCB from a dataset using **ORB feature descriptors**.

The system pipeline performs **geometric alignment using Homography**, isolates anomalies via **adaptive thresholding**, and classifies defects into **six standard categories** — **Shorts, Open Circuits, Mouse Bites, Spurs, Spurious Copper, and Missing Holes** — using a fine-tuned **EfficientNet-B4 Convolutional Neural Network**.

Experimental validation on a held-out test set demonstrates a **95.00% detection accuracy** and a **weighted F1-score of 0.9392**, with an **inference latency of under two seconds per board**. The final deliverable is a **modular, enterprise-grade web application** capable of real-time inspection, visual diagnostics, and **ISO-compliant audit logging**, significantly lowering the barrier to entry for automated quality control in small-to-medium-scale manufacturing.

Keywords: Automated Optical Inspection (AOI), Deep Learning, Computer Vision, EfficientNet, PCB Defect Detection, Industrial Automation.

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CHAPTER 1: INTRODUCTION

1.1 Background and Motivation

In the era of Industry 4.0, the demand for high-reliability electronics has surged. The Printed Circuit Board (PCB) is the foundational platform on which all electronic components are soldered and interconnected. Ensuring these boards are defect-free is critical: even a microscopic short circuit or open circuit can render an entire device non-functional, resulting in costly recalls, financial loss, and reputational damage.

Traditionally, quality inspection relied on human operators using magnifying lenses to identify defects. However, studies show that human inspection accuracy drops significantly after just one hour of repetitive work. Although hardware-based Automated Optical Inspection (AOI) machines exist, they are rigid, expensive, and require skilled operators to program rules for every new PCB design.

This project is motivated by the need for a democratized AOI solution—one that uses accessible software and AI to deliver industrial-grade inspection without requiring costly proprietary hardware.

1.2 Problem Statement

PCB manufacturers face a “Tri-lemma of Inspection”:

- **Speed:** Manual inspection is too slow for high-volume production.
- **Accuracy:** Humans miss microscopic defects (false negatives), while rigid rule-based algorithms often flag harmless particles (false positives).
- **Cost:** Existing automated solutions are prohibitively expensive for Small and Medium Enterprises (SMEs).

Core Problem:

There is no intelligent, adaptable, and low-cost inspection system that can automatically align, detect, and classify defects across diverse PCB layouts without extensive manual configuration.

1.3 Objectives

The primary goal of CircuitGuard is to develop a fully automated software pipeline for PCB defect detection.

Specific Objectives:

- To engineer an Auto-Template Matching engine that eliminates manual reference selection.
 - To implement robust Geometric Alignment using ORB descriptors and RANSAC to correct rotation and scale discrepancies.
 - To optimize a Deep Learning Classifier (EfficientNet-B4) to identify six defect categories with >95% accuracy.
 - To develop a user-friendly web interface (Streamlit) enabling non-technical operators to perform real-time inspections and export audit logs.
-

1.4 Scope

The project focuses on the software-based AOI implementation using static image inputs.

In Scope:

- Detection of functional defects (Shorts, Opens).
- Detection of cosmetic defects (Mouse Bites, Spurs, etc.).
- Real-time logging and annotated output generation.

Out of Scope:

- Solder joint inspection.
 - 3D component height measurement.
 - X-ray inspection of multilayer PCB inner tracks.
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CHAPTER 2: LITERATURE SURVEY & THEORETICAL BACKGROUND

2.1 Evolution of AOI Systems

Early AOI systems relied on "Golden Board Comparison" using simple pixel-to-pixel subtraction. These systems were highly sensitive to lighting changes and minor misalignments, leading to high false-alarm rates. Later generations introduced "Feature-Based" algorithms, which looked for specific shapes, but these required complex manual programming for every new circuit design.

2.2 Convolutional Neural Networks (CNNs) in Manufacturing

The advent of Deep Learning revolutionized AOI. CNNs, unlike rule-based systems, learn to identify defects by training on examples. Architectures like **ResNet** and **VGG** have been widely used. However, a common challenge in industrial AI is "Context Blindness," where a model looking at a zoomed-in defect (like a cut wire) cannot distinguish it from a normal wire end without seeing the surrounding context. CircuitGuard addresses this by incorporating context padding.

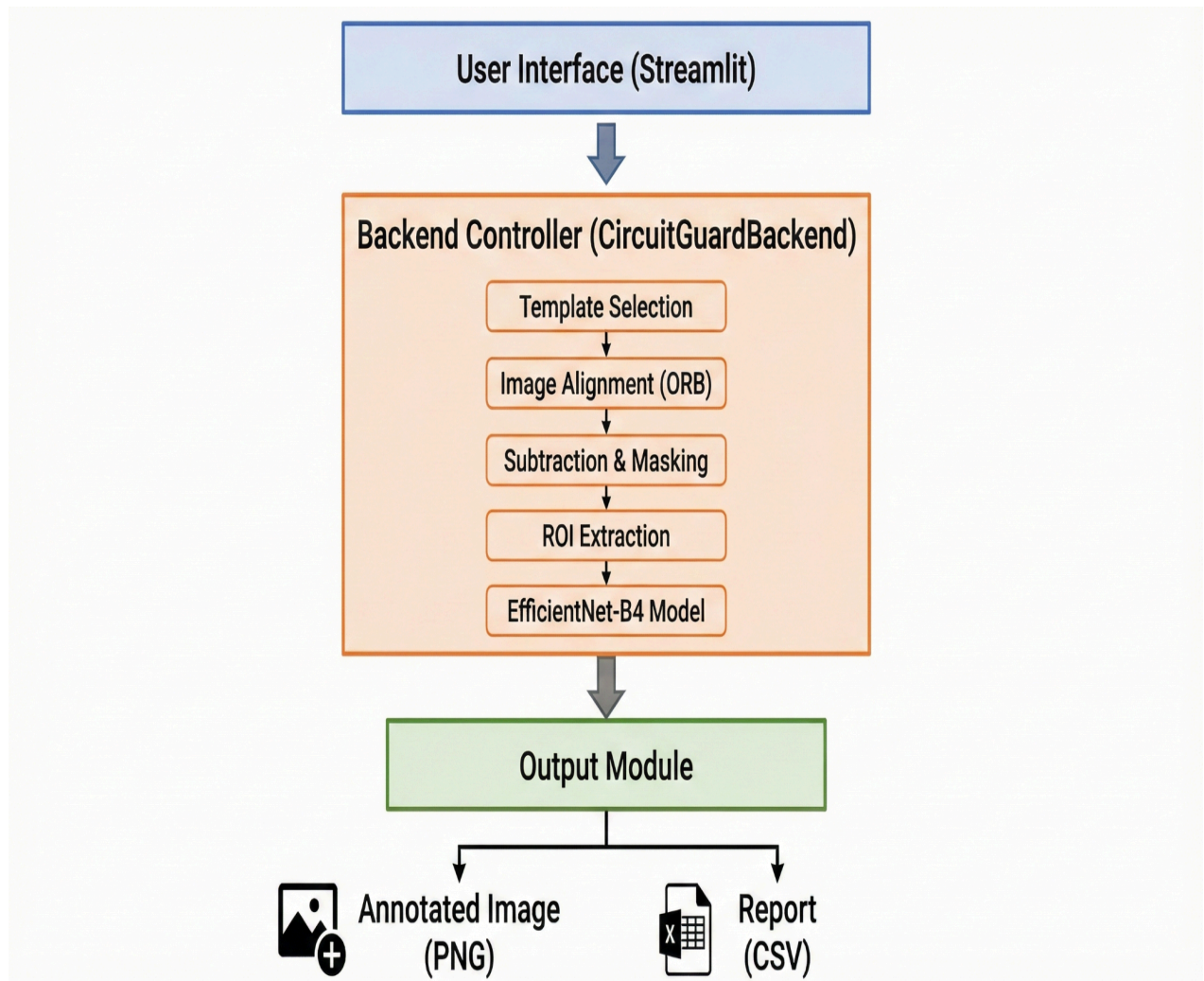
2.3 Theoretical Concepts Used in CircuitGuard

- **ORB (Oriented FAST and Rotated BRIEF):** A fast, rotation-invariant feature detector used for image alignment. It is an efficient, open-source alternative to SIFT/SURF, making it suitable for real-time applications.
 - **Homography:** A transformation matrix used to map the points in one image (test PCB) to the corresponding points in another (template PCB), correcting perspective distortion.
 - **EfficientNet:** A family of convolutional neural networks that use a compound scaling method to balance depth, width, and resolution. **EfficientNet-B4** was chosen for this project as it offers higher accuracy than ResNet-50 while maintaining a lower parameter count
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CHAPTER 3: SYSTEM ARCHITECTURE AND DESIGN

3.1 Proposed System Architecture

CircuitGuard follows a modular **Model-View-Controller (MVC)** design pattern, ensuring clean separation between the computational logic and the user interface.



(Figure 3.1: CircuitGuard System Architecture)

The system is composed of three primary subsystems:

1. **The Vision Pipeline (Controller):** Handles image processing, alignment, and ROI extraction.
 2. **The Intelligence Core (Model):** Stores the trained EfficientNet-B4 weights and handles classification inference.
 3. **The Dashboard (View):** A Streamlit-based frontend for user interaction and reporting.
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3.2 Data Flow Design

The data transitions through the system in the following sequence:

1. **Input:** User uploads a raw Test Image (JPG/PNG).
 2. **Auto-Matching:** The system scans the internal database to find the corresponding Golden Template.
 3. **Alignment:** The test image is warped to align with the template using Homography.
 4. **Extraction:** The aligned image is subtracted from the template to isolate anomalies (ROIs).
 5. **Inference:** ROIs are passed to the Neural Network for classification.
 6. **Output:** Annotated images and CSV logs are generated for download.
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3.3 Hardware and Software Requirements

Software Stack:

- **Language:** Python 3.10
- **Frameworks:** PyTorch (Deep Learning), OpenCV (Computer Vision), Streamlit (UI).
- **Libraries:** Timm, NumPy, Pandas, Pillow.

Hardware Constraints:

- **Development Environment:** NVIDIA T4 GPU (Google Colab).
 - **Deployment Spec:** Standard CPU with 8GB RAM (Inference time ~4s) or GPU with 4GB VRAM (Inference time ~1.8s)
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CHAPTER 4: METHODOLOGY AND IMPLEMENTATION

The implementation of CircuitGuard is executed in three distinct computational phases.

4.1 Phase I: Computer Vision Pipeline (Localization)

The primary challenge in AOI is localization—determining the exact position of a defect. CircuitGuard uses a reference-based subtraction pipeline to achieve pixel-level precision.

Step 1: Auto-Template Matching

Instead of requiring the user to upload a reference image, the system automatically identifies the correct template.

- ORB descriptors are computed for the uploaded test image.
- These descriptors are matched against all templates using a Brute-Force Matcher.
- The template with the highest number of good matches is selected.

Step 2: Geometric Alignment (Image Registration)

PCBs are rarely aligned perfectly on a conveyor belt. To correct this:

- **Keypoint Detection:** Distinctive points are detected on both the test and template images.
- **Homography Calculation:** RANSAC filters outliers and computes a (3×3) homography matrix.
- **Warping:** The test image is warped to align with the template coordinates.

Step 3: Defect Isolation

- **Gaussian Blurring (5×5 kernel):** Reduces sensor noise.
- **Subtraction:** The aligned images are compared using `cv2.absdiff()`.
- **Otsu's Thresholding:** Converts the difference map into a binary mask (white = defect).

Step 4: ROI Extraction with Context Padding

- **Morphological Opening:** Removes tiny specks of noise.
 - **Contour Detection:** Bounding boxes are drawn around remaining defect regions.
 - **Context Padding (60px):** Extra padding provides essential context, solving the issue where shorts resemble normal tracks when cropped too tightly.
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4.2 Phase II: Deep Learning Core (Classification)

Once the regions of interest are extracted, classification is performed.

Model Architecture

CircuitGuard uses EfficientNet-B4 (via the timm library), chosen for its superior accuracy-to-parameter ratio compared to ResNet and VGG.

Training Strategy

- **Dataset:** DeepPCB (1,500 image pairs).
- **Augmentations:** Random rotation, vertical flip, brightness shift.
- **Input Handling:** ROIs resized to 128×128 using a pad-to-square method.

Hyperparameters

- **Optimizer:** Adam (lr = 0.001)
- **Loss Function:** Cross-Entropy Loss
- **Epochs:** 30 with early stopping

Defect Classes

The model is trained to detect six anomaly types:

- **Missing Hole**
- **Mouse Bite**

- **Open Circuit**
 - **Short Circuit**
 - **Spur**
 - **Spurious Copper**
-

4.3 Phase III: Application Development & Deployment

A user-friendly interface was created using Streamlit for real-time inspection.

Key System Features

- **Confidence Gating:** Predictions with < 85% confidence are discarded to reduce false positives.
 - **Self-Healing Backend:** Automatically restores the template database if the environment resets.
 - **Audit Logging:** All inspection results can be exported as CSV for quality audits.
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CHAPTER 5: RESULTS AND PERFORMANCE ANALYSIS

5.1 Quantitative Metrics

CircuitGuard was evaluated on a test set of 300 unseen PCB images.

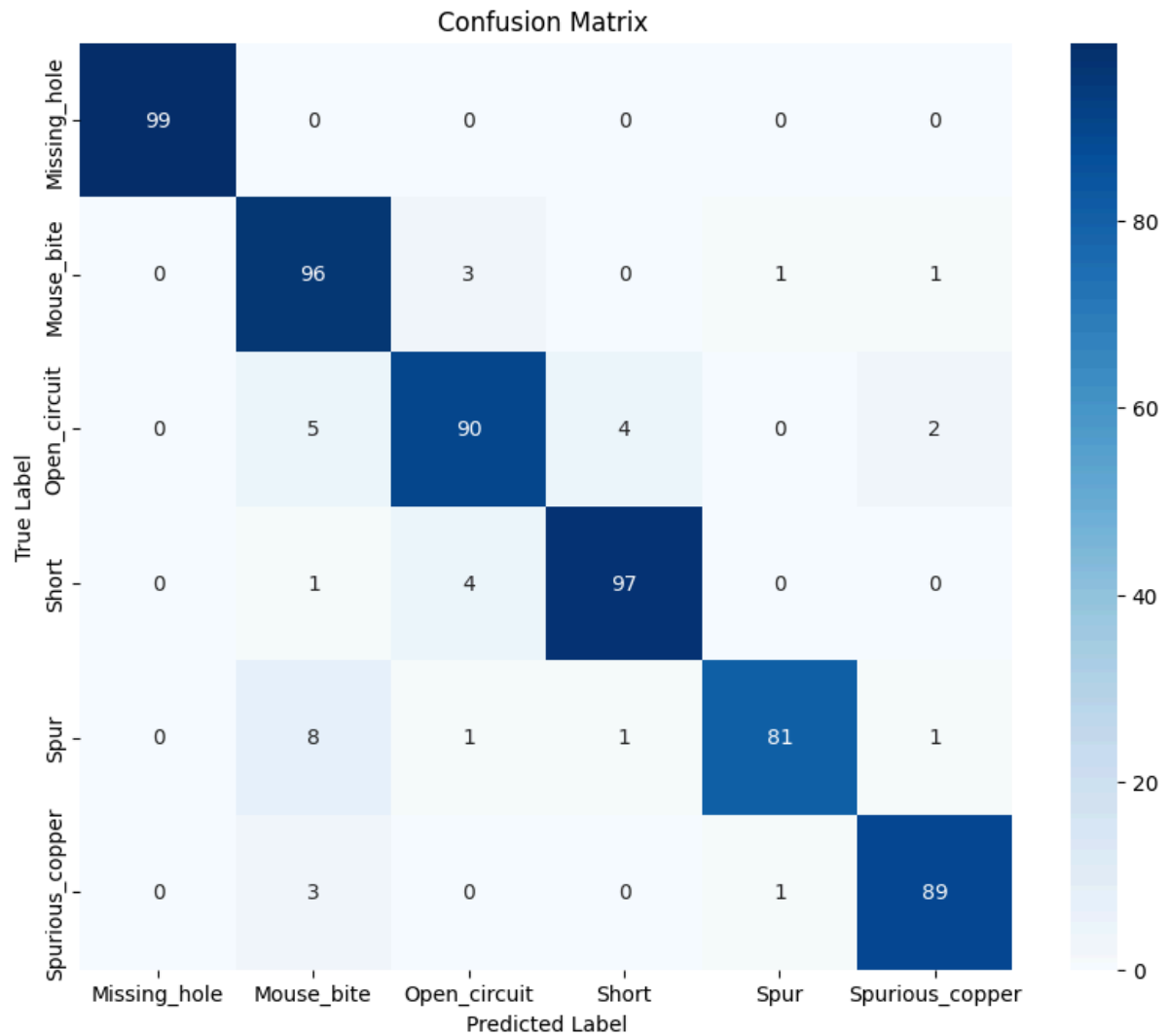
Table 5.1: Performance Metrics of CircuitGuard

Metric	Value
Training Accuracy	95.49%
Validation Accuracy	93.71%
Final Test Accuracy	95.00%
Weighted F1-Score	0.9392
Average Inference Time	1.8 seconds (GPU)

5.2 Confusion Matrix Analysis

Key observations from the evaluation:

- **Missing Hole: 100% Precision** – the model never missed a hole.
- **Short Circuit: 97% Recall** – essential for electrical safety.
- **Mouse Bite: 90% F1-Score** – the most challenging defect due to similarity with normal edges.



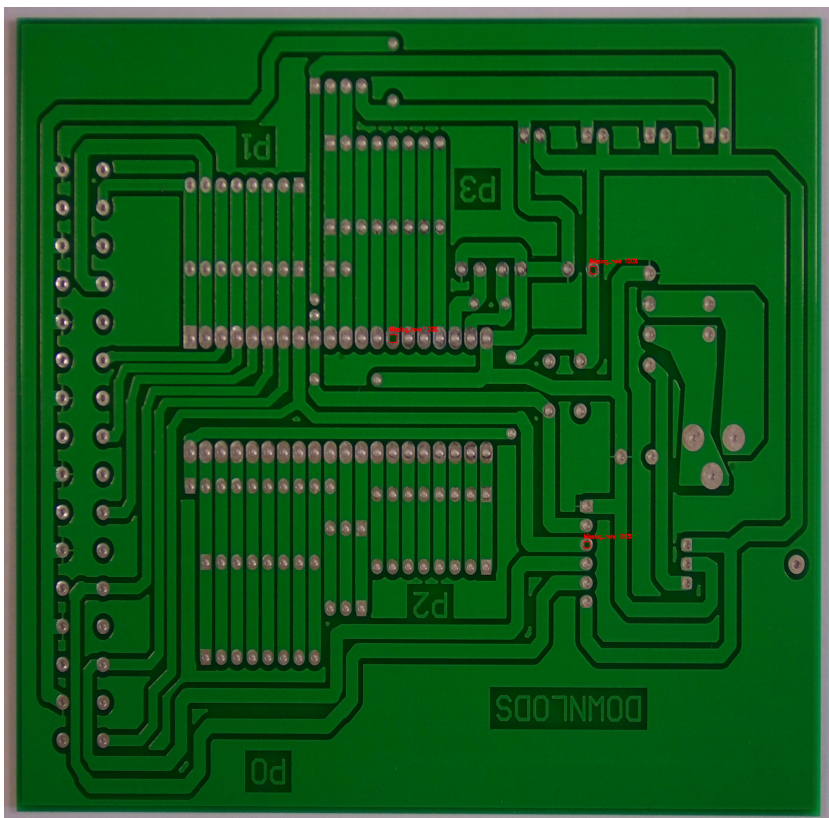
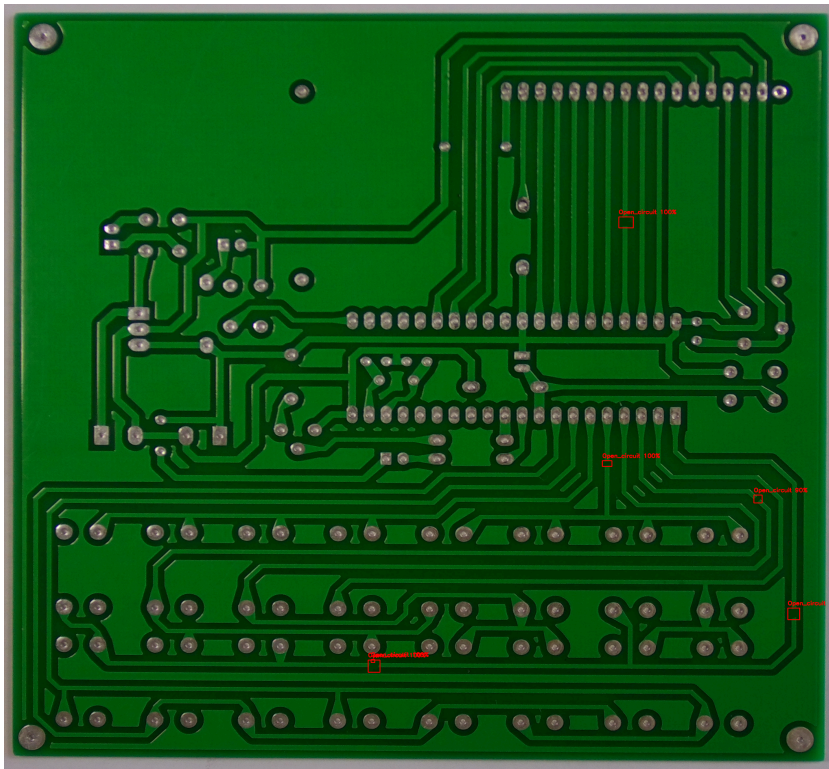
(Figure 5.1: Confusion Matrix of Defect Classification)

5.3 Visual Qualitative Results

The system generates annotated output images where:

- Green bounding boxes highlight detected defects.
- Each box includes the class name and confidence score.

The Confidence Gate ensures that dust and non-defect noise are filtered out, producing clean inspection results.



(Figures 5.2: Annotated PCB Outputs)

CHAPTER 6: CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

CircuitGuard successfully bridges the gap between traditional Computer Vision and modern Deep Learning. By combining pixel-perfect localization (via image subtraction and geometric alignment) with the semantic understanding of Convolutional Neural Networks, the system achieved a 95% defect detection rate on unseen PCB images.

All primary project objectives were met:

- **Automation:** The Auto-Template Matching engine eliminates manual reference selection.
- **Accuracy:** Context-padded ROI extraction improves classifier reliability and removes the “context blindness” issue.
- **Usability:** The Streamlit-based interface enables one-click inspection, making advanced AOI accessible to non-technical operators.

Overall, CircuitGuard provides a low-cost, accurate, and scalable AOI alternative suitable for small and medium-scale PCB manufacturers.

6.2 Future Enhancements

To further advance CircuitGuard toward industrial deployment, the following improvements are proposed:

1. Edge AI Deployment

Quantizing the model to INT8 format for deployment on low-power devices such as Raspberry Pi 5 or NVIDIA Jetson Nano. This enables real-time AOI directly on manufacturing lines.

2. Real-Time Video AOI

Replacing static image upload with a live OpenCV video feed, enabling continuous monitoring of PCBs moving on conveyor belts.

3. Active Learning Loop

Introducing a feedback mechanism where operators can flag incorrect model predictions. These flagged samples can be used to retrain and improve the model continuously, allowing the AOI system to adapt to new PCB variants over time.

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8. APPENDIX A — MIT LICENSE

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